



DESIGN OPTIMIZATION OF THE INSULATION DOMAIN IN A BUILDING WALL

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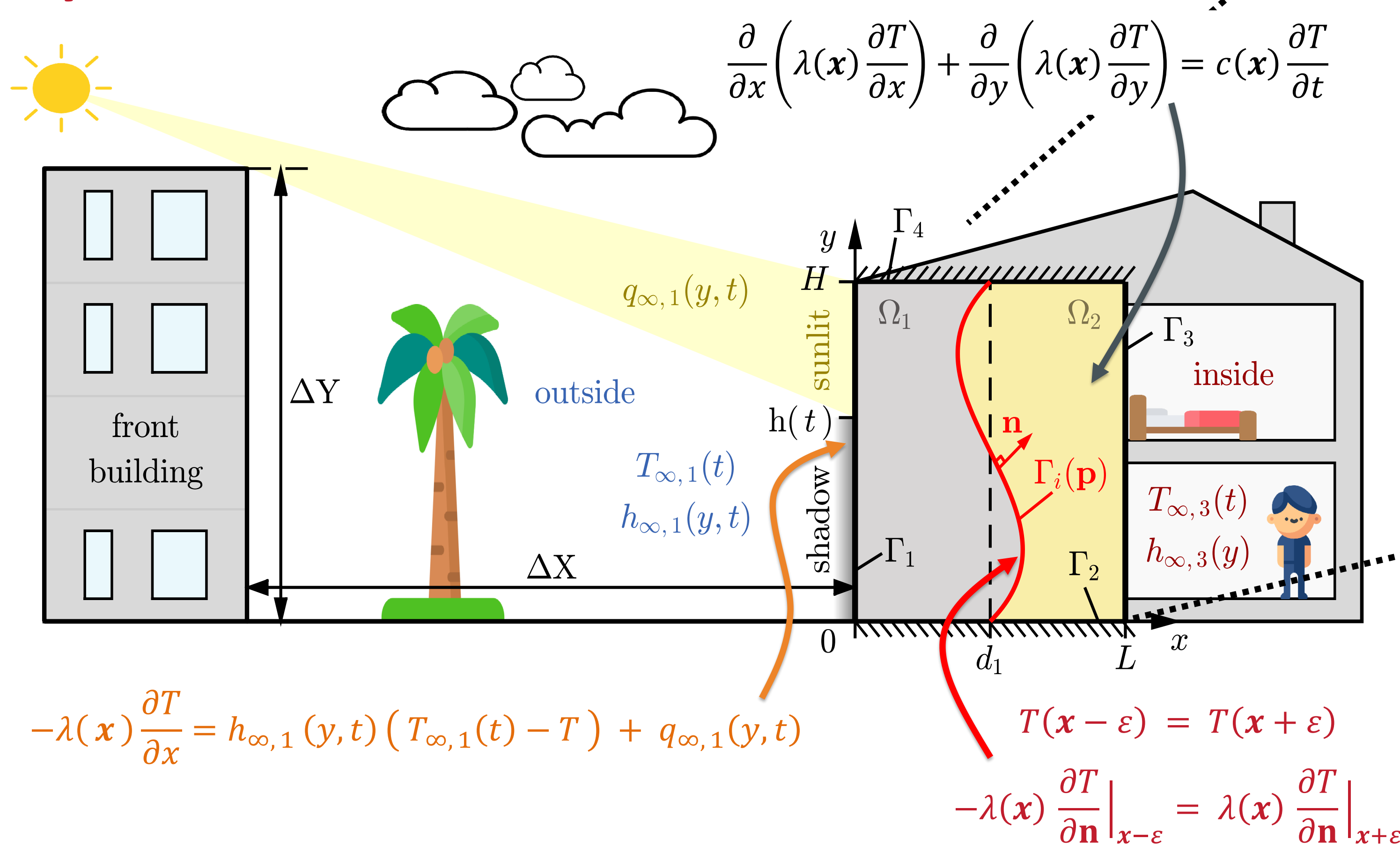
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Introduction

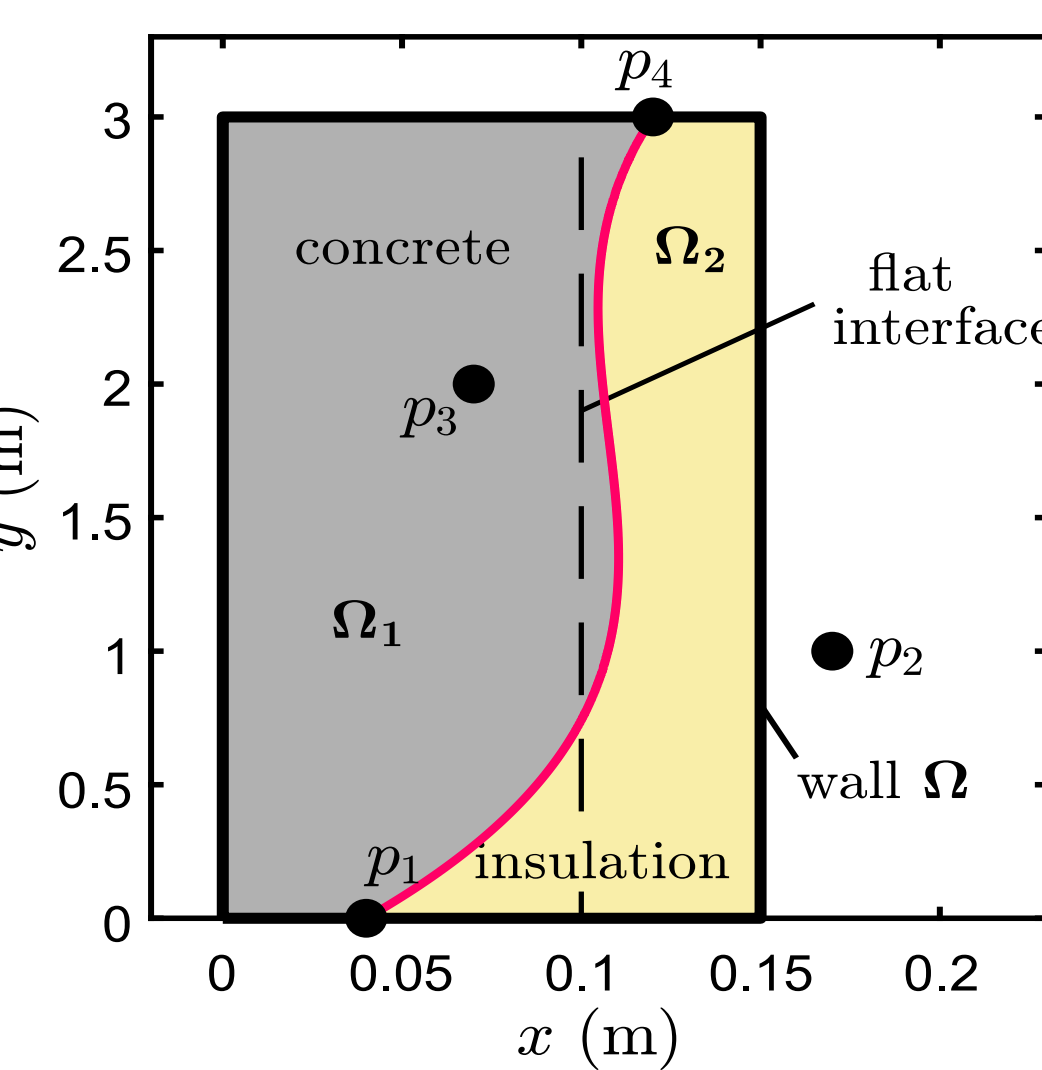
- Building envelopes are subject to complex thermal loads: solar irradiation, wind-dependent convective coefficients.
- Conventional models rely on simplifying assumptions [1], which limit their ability to capture these boundary heterogeneities.
- New opportunities:** additive manufacturing in construction enable complex geometries [2].
- Research gap:** optimization studies mostly focus on steady-state analyses [3], neglecting transient environmental conditions.

Objective: This work presents a novel optimization framework for thermal wall design by geometrically optimizing the interface between structural and insulating materials. The objective is to minimize energy demand.

Physical model



— interface • particles



Interface definition

The interface Γ_i is represented using a **Bézier** curve defined with 4 control points and Bernstein polynomial interpolation:

$$f_{B_x}(s, \mathbf{p}) = \sum_{i=1}^N \mathcal{B}_{i-1}^N(s) \cdot x_i + A(\mathbf{p})$$

$$f_{B_y}(s, \mathbf{p}) = \sum_{i=1}^N \mathcal{B}_{i-1}^N(s) \cdot y_i, \quad \forall s \in [0, 1]$$

Case study in Brazil

- Promising performance** tested under conditions of low internal insulation.
- The optimal interface in Rio de Janeiro **redistributes and reduces** the average **internal heat flux** $\bar{q}_3$ across the entire height of the wall, and **reduces** the magnitude of the average **external heat flux** $\bar{q}_1$ compared to a flat interface.

Thermal design problem

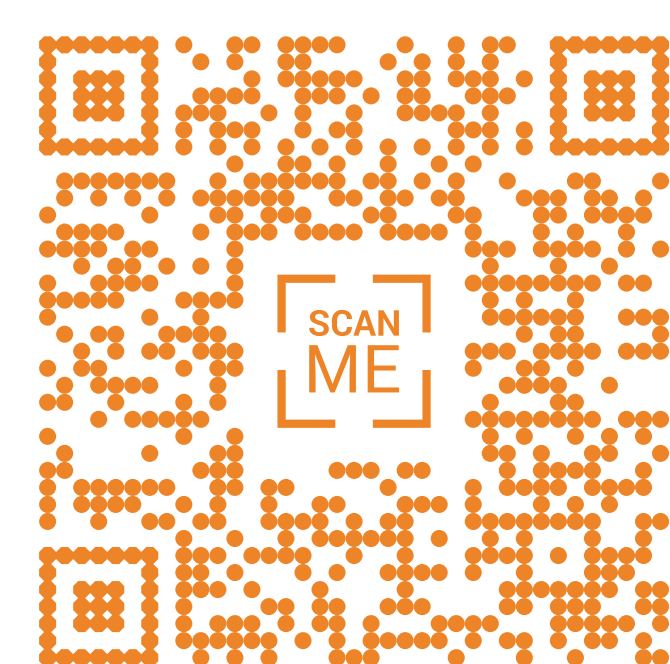
The **minimization problem** aims to find the optimal coordinates of the control points $\mathbf{p}^* = \min_{\mathbf{p} \in \Omega_p} \mathcal{C}$ of the Bézier curve, defined as:

$$\mathcal{C} = \int_0^{t_f} w(t) dt, \quad \text{with } w(t) = \left(\int_{\Gamma_3} -\lambda(\mathbf{x}) \cdot \frac{\partial T}{\partial x} dy \right) \cdot \left(\frac{T_{\infty,1}(t)}{T_{\infty,3}(t)} - 1 \right)$$

subject to: **Volume constraint**,
Spatial constraint.

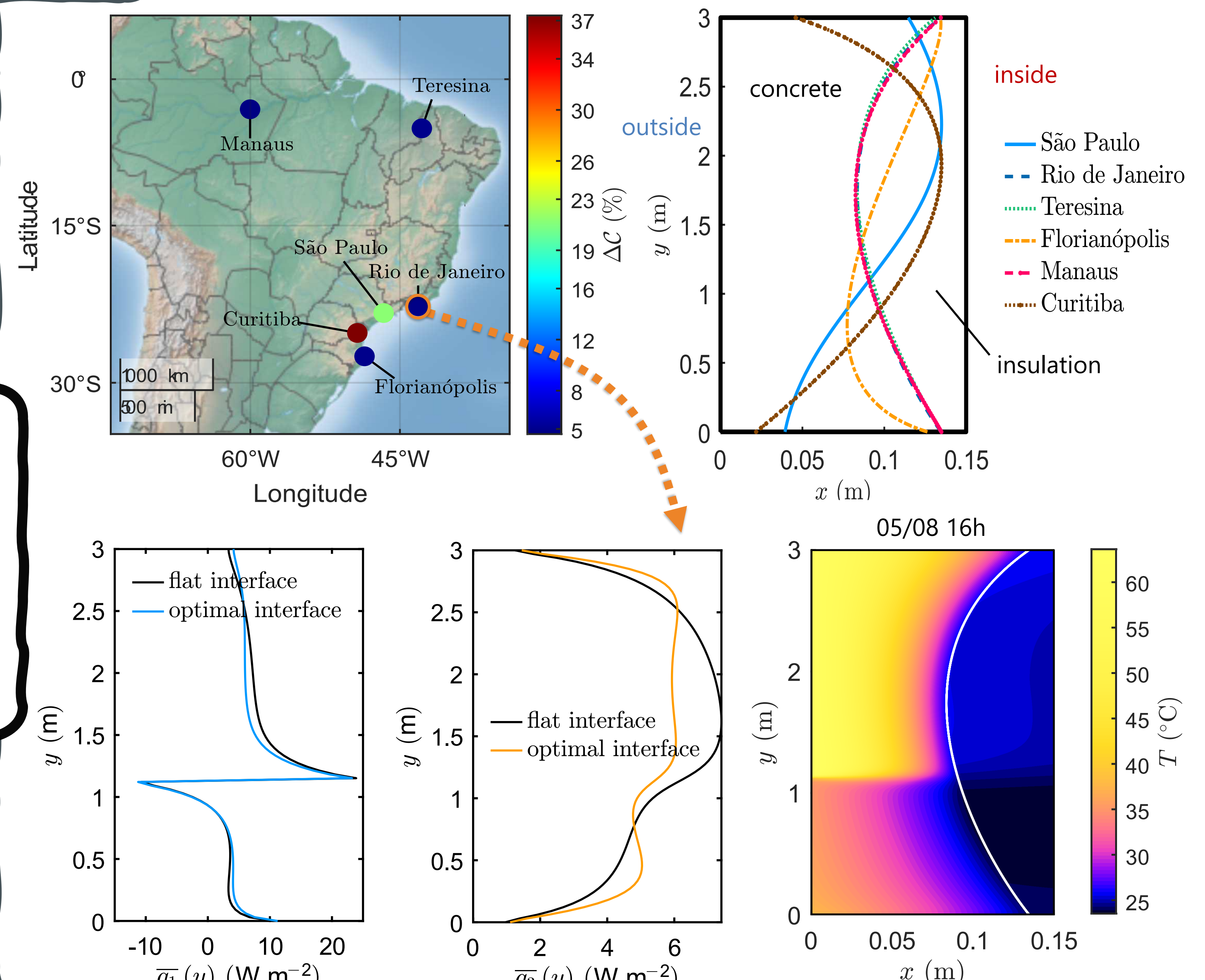
NSGA-II convergence assessment = relative objective variation $\delta \mathcal{C}$ + mean Euclidean parameter dispersion

Poster is animated !



Conclusion & perspectives

Despite geometric limitations (Bézier curves with $N_c = 4$ control points) and the absence of building-scale thermal simulations, annual energy gains of up to nearly 40% are achieved for the north-oriented façade compared with flat interfaces, highlighting the **potential of interface geometry optimization**.



References

- [1] M. Woloszyn and C. Rode. *Tools for performance simulation of heat, air and moisture conditions of whole buildings*. Building Simulation, 2008.
- [2] G. Vantghem et al., *Density-based topology optimization for 3D-printable building structures*, Structural and Multidisciplinary Optimization, 2019.
- [3] Z. Karashbayeva et al., *Optimization of the insulation domain for heat transfer in building walls*, Journal of the Brazilian Society of Mechanical Sciences and Engineering, 2025.

Acknowledgements

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